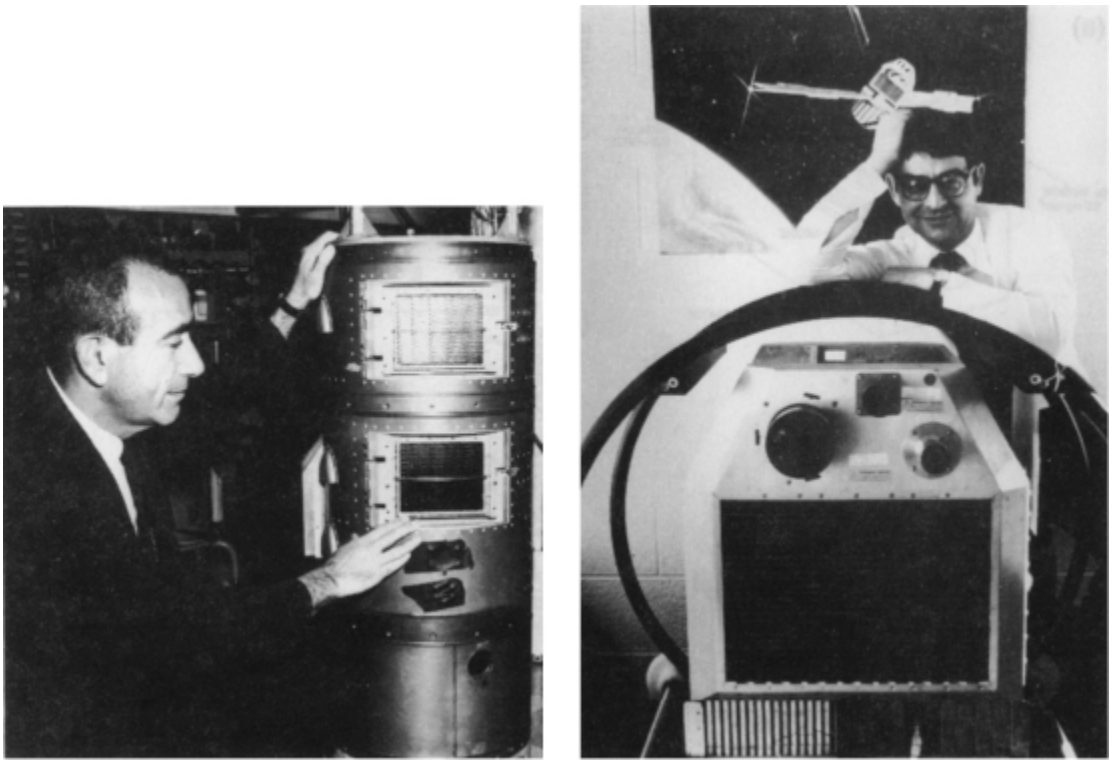


Left: **Herbert Friedman, with payload from an Aerobee rocket, in 1968.** ***Right:*** **Riccardo Giacconi with the Uhuru X-ray detector, ca. 1970.** [Left: courtesy U.S. Naval Research Laboratory; right: courtesy R. Giacconi.]

Giacconi's seasoned team took its first astronomical step with a search for X-rays from the Moon, using a detector patterned after Friedman's, and like Friedman, flying it on an Aerobee rocket. Their rocket, launched from White Sands, New Mexico, at one minute before midnight on 18 June 1962, climbed quickly to an altitude of 230 kilometers, then fell back to Earth. For 350 seconds it was high enough above the Earth's atmosphere to detect the Moon's X-rays. The data, telemetered back to the ground, were puzzling; the X-rays were far stronger than expected. When examined more closely, the data were even more surprising. The X-rays seemed to be coming not from the Moon, but from the constellation Scorpius (Figure 8.6b). For two months, Giacconi and his team members (Herbert Gursky, Frank Paolini, and Bruno Rossi) sought errors in their data and apparatus. When none could be found, they announced their discovery: The first X-ray star ever detected, *5000 times brighter than theoretical astrophysicists had predicted* Ten months later, Friedman's team confirmed the discovery, and the star was given the name Sco X-1 (*1* for "the brightest," *X* for "X-ray source," *Sco* for "in the constellation Scorpius").